R. PRAVEEN KUMAR

Industrial IoT & Software Developer | C++ · Qt/QML · C# · Python

pravinpforpolite@gmail.com | +91-9500214154 | Chengalpattu, Tamil Nadu, India

LinkedIn | GitHub

PROFESSIONAL SUMMARY

Software developer building Industrial IoT and edge applications in **C++, C#, Python and Qt/QML** for real-time manufacturing environments. Over 2 years of dedicated IIoT development, including a year in Nissan Yokohama, Japan, covering CNC/PLC data acquisition (FANUC FOCAS, Keyence KVCom+), HMI and dashboard development, and computer-vision error-proofing with OpenCV and YOLO. Certified F-Pilot developer working to Nissan software standards: architecture, implementation, unit/integration testing, defect investigation and performance tuning. A mechanical engineering background (12+ years in powertrain manufacturing) gives me deep domain knowledge of the machines and processes my software runs on.

TECHNICAL SKILLS

Programming Languages: C++ (C++17 and above), Embedded C/C++, C#, Python, VB, MATLAB, Excel VBA

Embedded & RTOS: Modern Embedded C++, FreeRTOS

GUI & HMI: Qt/QML, PySimpleGUI; machine diagnostic screens and operator interfaces

IoT, Edge & Protocols: FANUC CNC (FOCAS), Keyence PLC (KVCom+), NC/PLC/PMC communication, edge computing, device SDK integration, factory networks

Computer Vision & AI: OpenCV, Ultralytics YOLO, Pandas, Matplotlib; vision-based Poka-Yoke

Build & Tooling: CMake, qmake, Visual Studio; MS Access

Industrial Automation: Siemens PLC (Step 7, Ladder Logic), Delta PLC, WinCC SCADA

Practices: Software architecture design, unit and integration testing, defect investigation, technical documentation

PROFESSIONAL EXPERIENCE

IoT Developer, Assistant Manager | RNTBCI, MWC, Chengalpattu Apr 2025 – Present

Real-time monitoring and smart-factory software | On-site

- Develop IIoT applications in **C++, C#, Python and VB** for NC/PLC/PMC data logging and real-time production monitoring.
- Build **edge-computing systems** and design **Qt/QML HMIs** for machine diagnostics and operator interfaces.
- Develop **vision-based Poka-Yoke solutions** using OpenCV and YOLO/Ultralytics, integrated with shop-floor machines through OEM SDKs for alerts and error-proofing.
- Certified to develop and maintain F-Pilot systems under Nissan development standards: software architecture, implementation, unit and integration testing, defect investigation and performance improvement.
- Own IoT edge-device deployment end to end: RFQ, specification, installation and shop-floor integration.
- Create dashboards and analytical tools that support production decision-making; lead plant-wide digital transformation initiatives.

IoT Developer, NPTP Trainee | Nissan Motor Corporation, Yokohama, Japan Apr 2024 – Mar 2025

Nissan Production Technology Partnership (NPTP) programme | On-site

- Selected for NPTP to specialise in Industrial IoT application development.
- Designed and deployed a **Quality Gate IoT application** for a new-technology machine, improving inspection accuracy and cycle time.
- Built CNC/PLC data-acquisition and communication solutions using FANUC FOCAS and Keyence KVCom+ over factory networks.
- Earned **Business Digital Engineer (BDE)** certification for deployment of digital process improvements.

IoT Data Analyst & RPA Developer, Senior Production Engineer | RNTBCI, MWC May 2021 – Mar 2024

Hybrid / On-site

- Built and deployed the **Python XIGMA Dashboard** at Nissan Yokohama; also built a Tooling Drawing Management System (VBA + MS Access), Excel-VBA tools and PySimpleGUI applications for tooling standardisation.
- Developed **MATLAB-based** automated error-judgment and real-time alert systems for NC machining during a 3-month on-site assignment in Japan.
- Applied data analysis and algorithm-based prediction to reduce operational failures and improve machining quality; supported Nissan Japan and global plants on digital transformation.

SELECTED PROJECTS & RECOGNITION

- **XIGMA Python Dashboard (Nissan Yokohama):** named Individual Achiever of the Month by the Department VP.
- **Quality Gate IoT Application (Japan):** developed and commissioned; earned BDE certification.
- **Automated Error-Judgment Vision System (Japan):** improved defect-detection accuracy at deployment.
- **BR Engine BS IV to BS VI Step-1 Upgrade:** Gold Special Award from the RNTBCI Managing Director.

CERTIFICATIONS

F-Pilot Crew System Developer | Nissan Motor Corporation | ID: UP2-F-2026-002 Oct 2026

Modern Embedded C++ with FreeRTOS | Future Shock Technologies | ID: FST-MECPP-001 May 2026

Business Digital Engineer (BDE) | Nissan Motor Corporation 2025

Course on PLC and SCADA | LIVEWIRE India | ID: LWR200229014 Feb 2020

EARLIER CAREER (MANUFACTURING ENGINEERING)

Engine Assembly & Testing Process Engineer | RNTBCI Apr 2017 – May 2021

- Designed assembly and test processes for prototype engines, including line balancing and layout; built wiring harnesses for engine test beds and ran RFQ, vendor selection and machine proving for mass-production readiness.

Project Management Engineer, Powertrain Projects | RNTBCI Sept 2014 – Mar 2017

- Handled CAPEX/OPEX budgeting, schedules and supplier coordination for powertrain programmes.

EDUCATION

B.E. Mechanical Engineering | KCG College of Technology, Chennai | CGPA 7.71 2010 – 2014

Languages: Tamil (native), English (professional), Japanese (basic conversational; currently studying for JLPT N5)